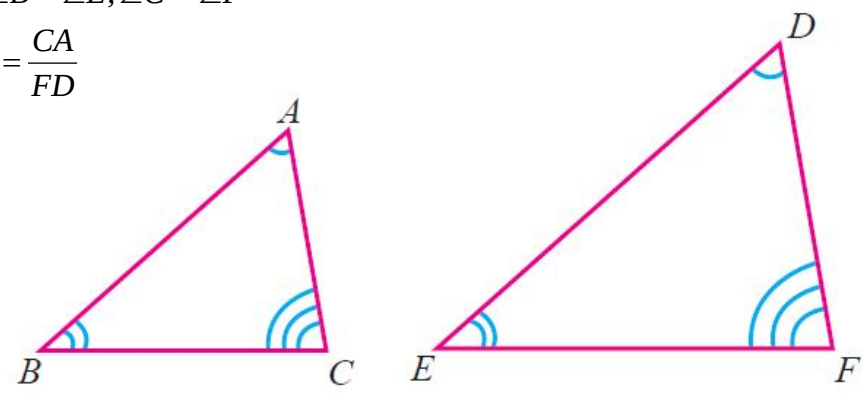
CLASS X : CHAPTER - 6
TRIANGLES

IMPORTANT FORMULAS & CONCEPTS

All those objects which have the same shape but different sizes are called similar objects.
Two triangles are similar if
(i) their corresponding angles are equal (or)
(ii) their corresponding sides have lengths in the same ratio (or proportional)

Two triangles ΔABC and ΔDEF are similar if
$(i) \angle A = \angle D, \angle B = \angle E, \angle C = \angle F$
$(ii) \frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}$



Basic Proportionality theorem or Thales Theorem
If a straight line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.

If in a ΔABC, a straight line DE parallel to BC, intersects AB at D and AC at E, then
$(i) \frac{AB}{AD} = \frac{AC}{AE}$ $(ii) \frac{AB}{DB} = \frac{AC}{EC}$

Converse of Basic Proportionality Theorem (Converse of Thales Theorem)
If a straight line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

Angle Bisector Theorem
The internal (external) bisector of an angle of a triangle divides the opposite side internally (externally) in the ratio of the corresponding sides containing the angle.

Converse of Angle Bisector Theorem
If a straight line through one vertex of a triangle divides the opposite side internally (externally) in the ratio of the other two sides, then the line bisects the angle internally (externally) at the vertex.

Criteria for similarity of triangles
The following three criteria are sufficient to prove that two triangles are similar.

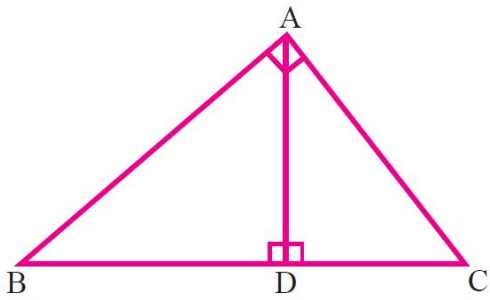
(i) AAA(Angle-Angle-Angle) similarity criterion
If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio (or proportion) and hence the two triangles are similar.
Remark: If two angles of one triangle are respectively equal to two angles of another triangle, then the two triangles are similar.

(ii) SSS (Side-Side-Side) similarity criterion for Two Triangles
In two triangles, if the sides of one triangle are proportional (in the same ratio) to the sides of the other triangle, then their corresponding angles are equal and hence the two triangles are similar.

(iii) SAS (Side-Angle-Side) similarity criterion for Two Triangles
If one angle of a triangle is equal to one angle of the other triangle and if the corresponding sides including these angles are proportional, then the two triangles are similar.

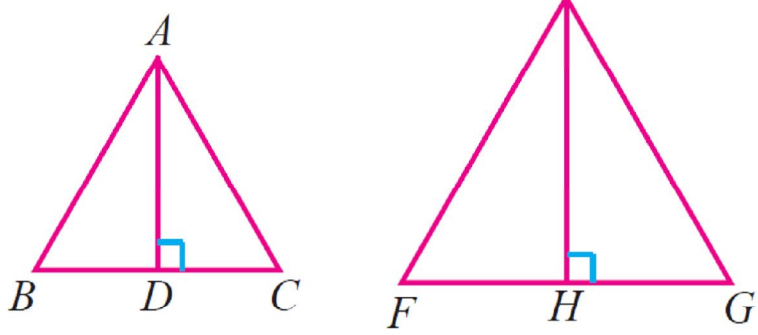
Areas of Similar Triangles
The ratio of the areas of two similar triangles is equal to the ratio of the squares of their corresponding sides.

If a perpendicular is drawn from the vertex of a right angled triangle to its hypotenuse, then the triangles on each side of the perpendicular are similar to the whole triangle.
Here, (a) $\Delta DBA + \Delta ABC$
(b) $\Delta DAC + \Delta ABC$
(c) $\Delta DBA + \Delta DAC$



If two triangles are similar, then the ratio of the corresponding sides is equal to the ratio of their corresponding altitudes.

i.e., if $\Delta ABC + \Delta EFG$, then $\frac{AB}{DE} = \frac{BC}{FG} = \frac{CA}{GE} = \frac{AD}{EH}$



If two triangles are similar, then the ratio of the corresponding sides is equal to the ratio of the corresponding perimeters.

If $\Delta ABC + \Delta EFG$, then $\frac{AB}{DE} = \frac{BC}{FG} = \frac{CA}{GE} = \frac{AB + BC + CA}{DE + FG + GE}$

Pythagoras theorem (Baudhayan theorem)
In a right angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

Converse of Pythagoras theorem
In a triangle, if the square of one side is equal to the sum of the squares of the other two sides, then the angle opposite to the first side is a right angle.

MCQ WORKSHEET-I
CLASS X : CHAPTER - 6
TRIANGLES

1. If in triangle ABC and DEF, $\frac{AB}{DE} = \frac{BC}{FD}$, then they will be similar when

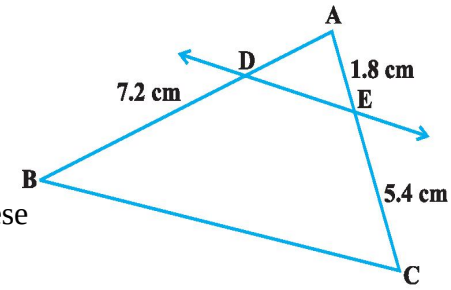
(a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$

2. It is given that $\Delta ABC \sim \Delta PQR$ with $\frac{BC}{QR} = \frac{1}{3}$, then $\frac{ar(\Delta ABC)}{ar(\Delta PQR)}$ is equal to

(a) 9 (b) 3 (c) $\frac{1}{3}$ (d) $\frac{1}{9}$

3. In ΔABC, DE || BC and AD = 4cm, AB = 9cm. AC = 13.5 cm then the value of EC is

(a) 6 cm (b) 7.5 cm (c) 9 cm (d) none of these



4. In figure DE || BC then the value of AD is

(a) 2 cm (b) 2.4 cm (c) 3 cm (d) none of the above

5. ABC and BDE are two equilateral triangles such that $BD = \frac{2}{3}BC$. The ratio of the areas of triangles ABC and BDE are

(a) 2 : 3 (b) 3 : 2 (c) 4 : 9 (d) 9 : 4

6. A ladder is placed against a wall such that its foot is at distance of 2.5 m from the wall and its top reaches a window 6 m above the ground. The length of the ladder is

(a) 6.5 m (b) 7.5 m (c) 8.5 m (d) 9.5 m

7. If the corresponding sides of two similar triangles are in the ratio 4 : 9, then the areas of these triangles are in the ratio is

(a) 2 : 3 (b) 3 : 2 (c) 81 : 16 (d) 16 : 81

8. If $\Delta ABC \sim \Delta PQR$, BC = 8 cm and QR = 6 cm, the ratio of the areas of Δ ABC and Δ PQR is

(a) 8 : 6 (b) 6 : 8 (c) 64 : 36 (d) 9 : 16

9. If $\Delta ABC \sim \Delta PQR$, area of Δ ABC = 81cm^2, area of Δ PQR = 144cm^2 and QR = 6 cm, then length of BC is

(a) 4 cm (b) 4.5 cm (c) 9 cm (d) 12 cm

10. Sides of triangles are given below. Which of these is a right triangle?

(a) 7 cm, 5 cm, 24 cm (b) 34 cm, 30 cm, 16 cm
(c) 4 cm, 3 cm, 7 cm (d) 8 cm, 12 cm, 14 cm

11. If a ladder 10 m long reaches a window 8 m above the ground, then the distance of the foot of the ladder from the base of the wall is

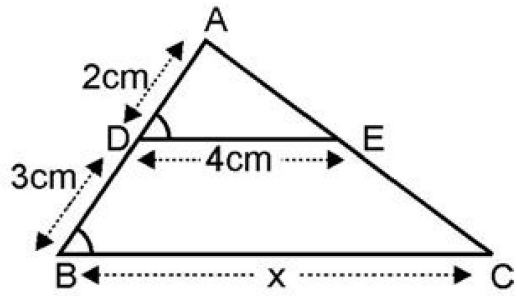
(a) 18 m (b) 8 m (c) 6 m (d) 4 m

12. A girl walks 200 towards East and the she walks 150m towards North. The distance of the girl from the starting point is

(a) 350m (b) 250m (c) 300m (d) 225m

MCQ WORKSHEET-II
CLASS X : CHAPTER - 6
TRIANGLES

1. In the given figure, if DE || BC, then x equals



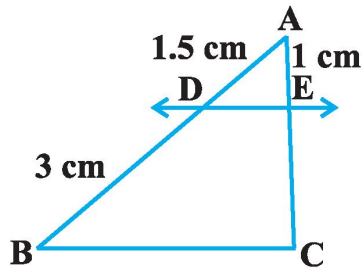
(a) 6 cm (b) 10 cm (c) 8 cm (d) 12.5 cm

2. All _____triangles are similar.
(a) isosceles (b) equilateral (c) scalene
(d) right angled

3. All circles are _____
(a) congruent (b) similar (c) not similar
(d) none of these

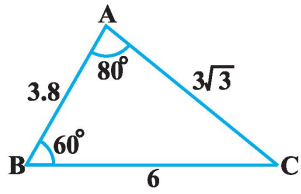
4. All squares are _____
(a) congruent (b) similar (c) not similar (d) none of these

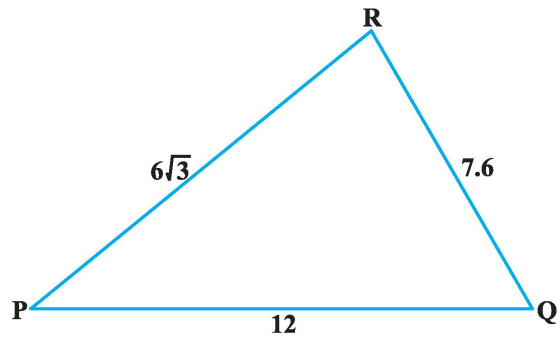
5. In the given fig DE || BC then the value of EC is



(a) 1 cm (b) 2 cm (c) 3 cm (d) 4 cm

6. In the given below figure, the value of $\angle P$ is





(a) 60^0 (b) 80^0 (c) 40^0 (d) 100^0

7. A girl of height 90 cm is walking away from the base of a lamp-post at a speed of 1.2 m/s. If the lamp is 3.6 m above the ground, then the length of her shadow after 4 seconds.
(a) 1.2 m (b) 1.6 m (c) 2 m (d) none of these

8. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.
(a) 42 m (b) 48 m (c) 54 m (d) none of these

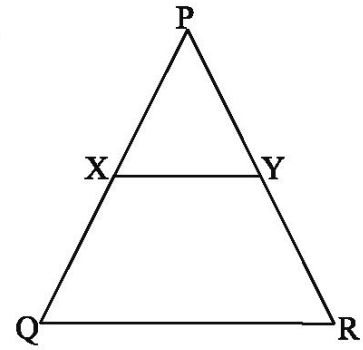
9. $\Delta ABC \sim \Delta DEF$ and their areas be, respectively, 64 cm^2 and 121 cm^2. If EF = 15.4 cm, the value of BC is.
(a) 11.2 cm (b) 15.4 cm (c) 6.4 cm (d) none of these

10. ABC and BDE are two equilateral triangles such that D is the midpoint of BC. Ratio of the areas of triangles ABC and BDE is
(a) 2 : 1 (b) 1 : 2 (c) 4 : 1 (d) 1 : 4

11. Areas of two similar triangles are in the ratio 4 : 9. Sides of these triangles are in the ratio
(a) 2 : 3 (b) 4 : 9 (c) 81 : 16 (d) 16 : 81

MCQ WORKSHEET-III
CLASS X : CHAPTER - 6
TRIANGLES

1. In the following fig. XY || QR and $\frac{PX}{XQ} = \frac{PY}{YR} = \frac{1}{2}$, then



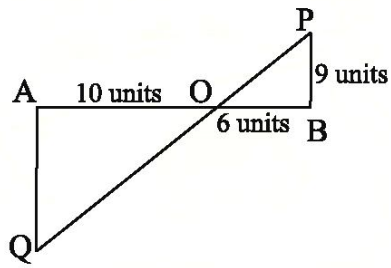
(*i*) XY = QR

(*ii*) $XY = \frac{1}{3}QR$

(*iii*) $XY^2 = QR^2$

(*iv*) $XY = \frac{1}{2}QR$

2. In the following fig QA ⊥ AB and PB ⊥ AB, then AQ is:



(i) 15 units (ii) 8 units (iii) 5 units (iv) 9 units

3. The ratio of the areas of two similar triangles is equal to the:

(i) ratio of their corresponding sides

(ii) ratio of their corresponding attitudes

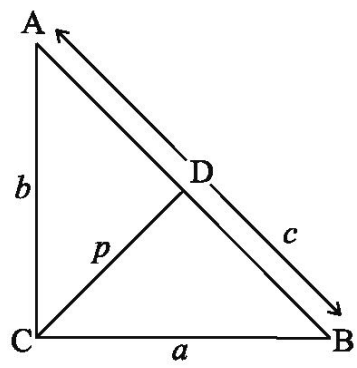
(iii) ratio of the squares of their corresponding sides

(iv) ratio of the squares of their perimeter

4. The areas of two similar triangles are 144 cm^2 and 81 cm^2. If one median of the first triangle is 16 cm, length of corresponding median of the second triangle is:

(i) 9 cm (ii) 27 cm (iii) 12 cm (iv) 16 cm

5. In a right triangle ABC, in which ∠C = 90° and CD ⊥AB. If BC = *a*, CA = *b*, AB = *c* and CD = *p*.



(*i*) $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

(*ii*) $\frac{1}{p^2} \neq \frac{1}{a^2} + \frac{1}{b^2}$

(*iii*) $\frac{1}{p^2} < \frac{1}{a^2} + \frac{1}{b^2}$

(*iv*) $\frac{1}{p^2} > \frac{1}{a^2} + \frac{1}{b^2}$

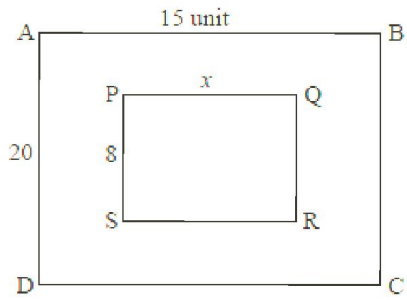
6. Given Quad. ABCD ~ Quad PQRS then x is:

(i) 13 units

(ii) 12 units

(iii) 6 units

(iv) 15 units



7. If $\Delta ABC \sim \Delta DEF$, ar($\Delta DEF$) = 100 cm^2 and AB/DE = 1/2 then ar(ΔABC) is:

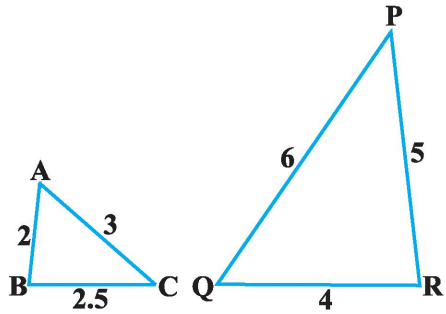
(i) 50 cm^2 (ii) 25 cm^2 (iii) 4 cm^2 (iv) None of the above.

8. If the three sides of a triangle are a, $\sqrt{3}a$, $\sqrt{2}a$, then the measure of the angle opposite to the longest side is:

(i) 45° (ii) 30° (iii) 60° (iv) 90°

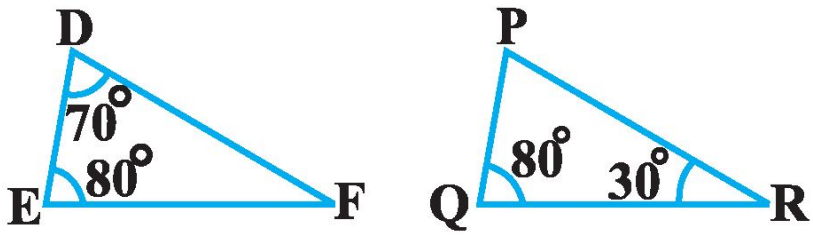
9. The similarity criterion used for the similarity of the given triangles shown in fig (iii) is

(a) AAA (b) SSS (c) SAS (d) AA



10. The similarity criterion used for the similarity of the given triangles shown in fig (iv) is

(a) AAA (b) SSS (c) SAS (d) AA



MCQ WORKSHEET-IV
CLASS X : CHAPTER - 6
TRIANGLES

1. A vertical pole of length 20 m casts a shadow 10 m long on the ground and at the same time a tower casts a shadow 50 m long, then the height of the tower.
(a) 100 m (b) 120 m (c) 25 m (d) none of these

2. The areas of two similar triangles are in the ratio 4 : 9. The corresponding sides of these triangles are in the ratio
(a) 2 : 3 (b) 4 : 9 (c) 81 : 16 (d) 16 : 81

3. The areas of two similar triangles ΔABC and ΔDEF are 144 cm^2 and 81 cm^2, respectively. If the longest side of larger ΔABC be 36 cm, then the longest side of the similar triangle ΔDEF is
(a) 20 cm (b) 26 cm (c) 27 cm (d) 30 cm

4. The areas of two similar triangles are in respectively 9 cm^2 and 16 cm^2. The ratio of their corresponding sides is
(a) 2 : 3 (b) 3 : 4 (c) 4 : 3 (d) 4 : 5

5. Two isosceles triangles have equal angles and their areas are in the ratio 16 : 25. The ratio of their corresponding heights is
(a) 3 : 2 (b) 5 : 4 (c) 5 : 7 (d) 4 : 5

6. If ΔABC and ΔDEF are similar such that 2AB = DE and BC = 8 cm, then EF =
(a) 16 cm (b) 112 cm (c) 8 cm (d) 4 cm

7. XY is drawn parallel to the base BC of a ΔABC cutting AB at X and AC at Y. If AB = 4BX and YC = 2 cm, then AY =
(a) 2 cm (b) 6 cm (c) 8 cm (d) 4 cm

8. Two poles of height 6 m and 11 m stand vertically upright on a plane ground. If the distance between their foot is 12 m, the distance between their tops is
(a) 14 cm (b) 12 cm (c) 13 cm (d) 11 cm

9. If D, E, F are midpoints of sides BC, CA and AB respectively of ΔABC, then the ratio of the areas of triangles DEF and ABC is
(a) 2 : 3 (b) 1 : 4 (c) 1 : 2 (d) 4 : 5

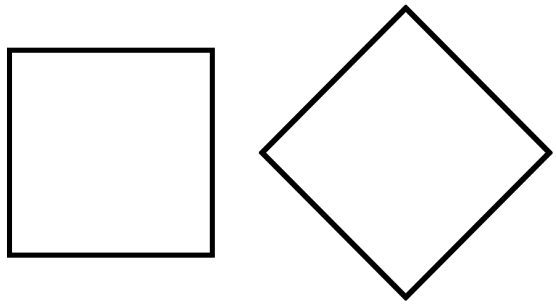
10. If ΔABC and ΔDEF are two triangles such that $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{2}{5}$, then $\frac{ar(\Delta ABC)}{ar(\Delta DEF)} =$
(a) 2 : 5 (b) 4 : 25 (c) 4 : 15 (d) 8 : 125

11. In triangles ABC and DEF, $\angle A = \angle E = 40^0$, AB : ED = AC : EF and $\angle F = 65^0$, then $\angle B =$
(a) 35^0 (b) 65^0 (c) 75^0 (d) 85^0

12. If ABC and DEF are similar triangles such that $\angle A = 47^0$ and $\angle E = 83^0$, then $\angle C =$
(a) 50^0 (b) 60^0 (c) 70^0 (d) 80^0

PRACTICE QUESTIONS
CLASS X : CHAPTER - 6
TRIANGLES

1. State whether the following pairs of polygons are similar or not:



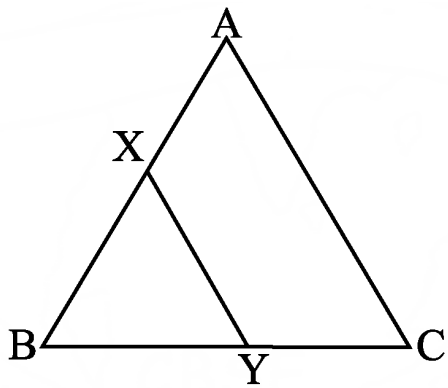
2. In triangle ABC, DE || BC and $\frac{AD}{DB} = \frac{3}{5}$. If AC = 4.8 cm, find AE.

3. A girl of height 90 cm is walking away from the base of a lamp post at a speed of 1.2m/s. If the lamp is 3.6 m above the ground, find the length of her shadow after 4 seconds.

4. Diagonals of a trapezium ABCD with AB || CD intersects at O. If AB = 2CD, find the ratio of areas of triangles AOB and COD.

5. Prove that the areas of two similar triangles are in the ratio of squares of their corresponding altitudes.

6. In the below figure, the line segment XY is parallel to side AC of ΔABC and it divides the triangle into two equal parts of equal areas. Find the ratio $\frac{AX}{AB}$.



7. In a triangle, if square of one side is equal to the sum of the squares of the other two sides, then the angle opposite the first side is a right angle. Prove it.

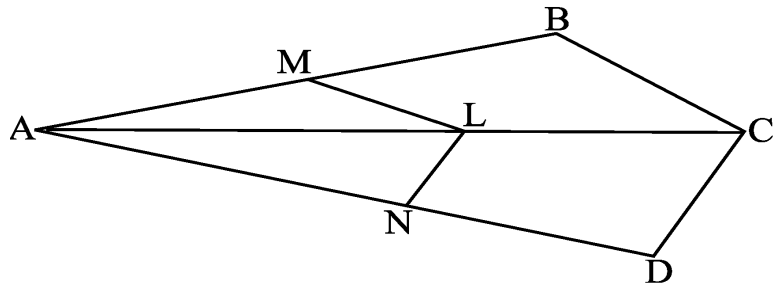
8. E is a point on the side AD produced of a ||gm ABCD and BE intersects CD at F. Show that ΔABE ~ ΔCFB.

9. Complete the sentence: Two polygons of the same number of sides are similar if…….

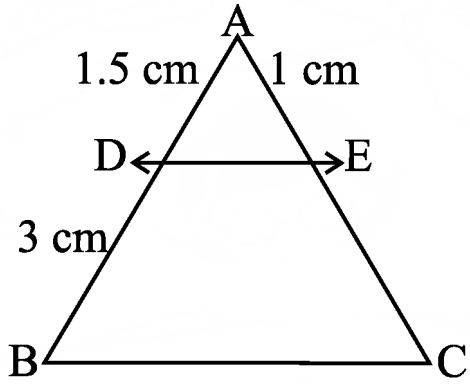
10. In ΔABC, AD ⊥ BC. Prove that $AB^2 - BD^2 = AC^2 - CD^2$.

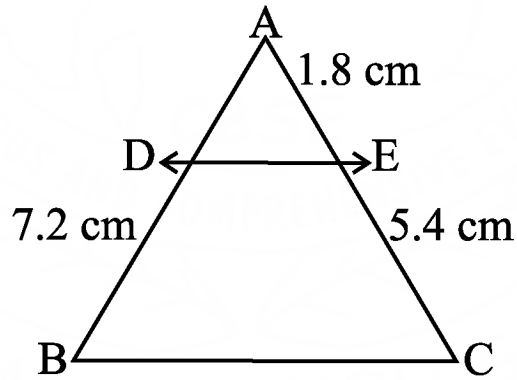
11. AD is a median of ΔABC. The bisector of ∠ADB and ∠ADC meet AB and AC in E and F respectively. Prove that EF || BC.

12. State and prove the Basic Proportionality theorem. In the below figure, if LM || CB and LN || CD, prove that $\frac{AM}{AB} = \frac{AN}{AD}$.



13. In the below figure, DE || BC, find EC





14. In the above right sided figure, DE || BC, find AD.

15. In given figure $\frac{AD}{DB} = \frac{AE}{EC}$ and $\angle AED = \angle ABC$. Show that AB = AC

16. $\Delta ABC \sim \Delta DEF$, such that ar($\Delta ABC$) = 64 cm^2 and ar(ΔDEF) = 121 cm2. If EF = 15.4 cm, find BC.

17. ABC and BDE are two equilateral triangles such that D is the midpoint of BC. What is the ratio of the areas of triangles ABC and BDE.

18. Sides of 2 similar triangles are in the ratio 4 : 9. What is the ratio areas of these triangles.

19. Sides of a triangle are 7cm, 24 cm, 25 cm. Will it form a right triangle? Why or why not?

20. Find $\angle B$ in ΔABC , if AB = $6\sqrt{3}$ cm, AC = 12 cm and BC = 6 cm.

21. Prove that "If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio".

22. Prove that "If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side."

23. If a line intersects sides AB and AC of a ΔABC at D and E respectively and is parallel to BC, prove that $\frac{AD}{AB} = \frac{AE}{AC}$

24. ABCD is a trapezium in which AB || DC and its diagonals intersect each other at the point O. Show that $\frac{AO}{BO} = \frac{CO}{DO}$

25. Prove that "If in two triangles, corresponding angles are equal, then their corresponding sides are in the same ratio (or proportion) and hence the two triangles are similar.

26. Prove that "If one angle of a triangle is equal to one angle of the other triangle and the sides including these angles are proportional, then the two triangles are similar.

27. Prove that "If in two triangles, sides of one triangle are proportional to (i.e., in the same ratio of) the sides of the other triangle, then their corresponding angles are equal and hence the two triangles are similar.

28. D is a point on the side BC of a triangle ABC such that $\angle ADC = \angle BAC$. Show that $CA^2 = CB.CD$.

29. Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of ΔPQR. Show that $\Delta ABC \sim \Delta PQR$.

30. Sides AB and AC and median AD of a triangle ABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that $\Delta ABC \sim \Delta PQR$.

31. If AD and PM are medians of triangles ABC and PQR, respectively where $\Delta ABC \sim \Delta PQR$, prove that $\frac{AB}{PQ} = \frac{AD}{PM}$

32. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

33. Prove that "The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides."

34. If the areas of two similar triangles are equal, prove that they are congruent.

35. D, E and F are respectively the mid-points of sides AB, BC and CA of ΔABC. Find the ratio of the areas of ΔDEF and ΔABC.

36. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding medians.

37. Prove that the area of an equilateral triangle described on one side of a square is equal to half the area of the equilateral triangle described on one of its diagonals.

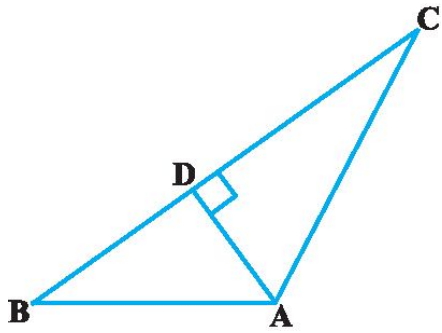
38. Prove that "If a perpendicular is drawn from the vertex of the right angle of a right triangle to the hypotenuse then triangles on both sides of the perpendicular are similar to the whole triangle and to each other."

39. Prove that "In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides.

40. O is any point inside a rectangle ABCD. Prove that $OB^2 + OD^2 = OA^2 + OC^2$.

41. Prove that the sum of the squares of the sides of a rhombus is equal to the sum of the squares of its diagonals.

42. In Fig., if AD ⊥ BC, prove that $AB^2 + CD^2 = BD^2 + AC^2$.



43. BL and CM are medians of a triangle ABC right angled at A. Prove that $4(BL^2 + CM^2) = 5\,BC^2$.

44. An aeroplane leaves an airport and flies due north at a speed of 1000 km per hour. At the same time, another aeroplane leaves the same airport and flies due west at a speed of 1200 km per hour. How far apart will be the two planes after $1^{1/2}$ hours?

45. D and E are points on the sides CA and CB respectively of a triangle ABC right angled at C. Prove that $AE^2 + BD^2 = AB^2 + DE^2$.

46. The perpendicular from A on side BC of a ΔABC intersects BC at D such that DB = 3 CD. Prove that $2AB^2 = 2AC^2 + BC^2$.

47. In an equilateral triangle ABC, D is a point on side BC such that BD = $\frac{1}{3}$BC. Prove that $9AD^2 = 7AB^2$.

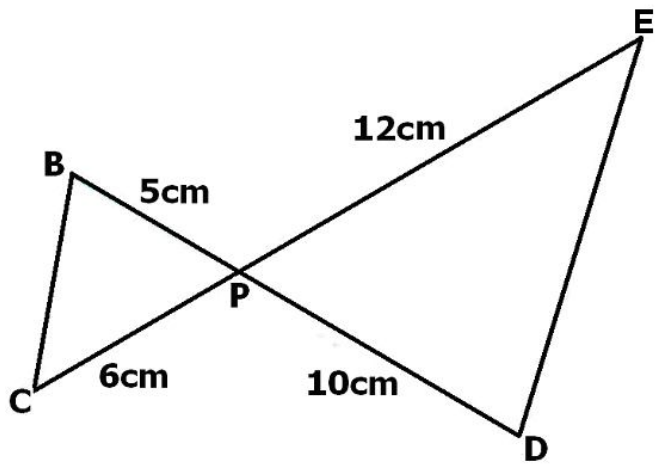
48. P and Q are the points on the sides DE and DF of a triangle DEF such that DP = 5 cm, DE = 15 cm, DQ= 6 cm and QF = 18 cm. Is PQ|| EF? Give reasons for your answer.

49. Is the triangle with sides 25 cm, 5 cm and 24 cm a right triangle? Give reasons for your answer.

50. It is given that Δ DEF ~ Δ RPQ. Is it true to say that ∠D = ∠R and ∠F = ∠P? Why?

51. A and B are respectively the points on the sides PQ and PR of a triangle PQR such that PQ = 12.5 cm, PA = 5 cm, BR= 6 cm and PB = 4 cm. Is AB || QR? Give reasons for your answer.

52. In the below Figure, BD and CE intersect each other at the point P. Is Δ PBC ~ ΔPDE? Why?



53. In triangles PQR and MST, ∠P = 55°, ∠Q = 25°, ∠M = 100° and ∠S = 25°. Is Δ QPR ~ Δ TSM? Why?

54. Is the following statement true? Why? "Two quadrilaterals are similar, if their corresponding angles are equal".

55. Two sides and the perimeter of one triangle are respectively three times the corresponding sides and the perimeter of the other triangle. Are the two triangles similar? Why?

56. If in two right triangles, one of the acute angles of one triangle is equal to an acute angle of the other triangle, can you say that the two triangles will be similar? Why?

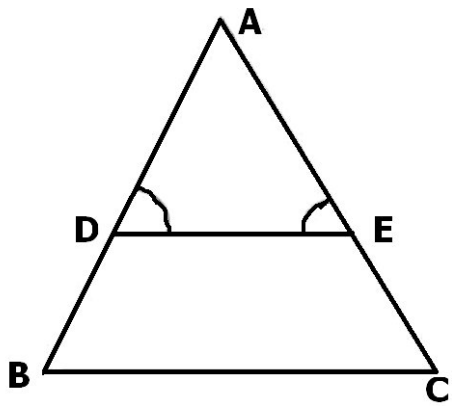
57. The ratio of the corresponding altitudes of two similar triangles is 3 : 5. Is it correct to say that ratio of their areas is 6 : 5 ? Why?

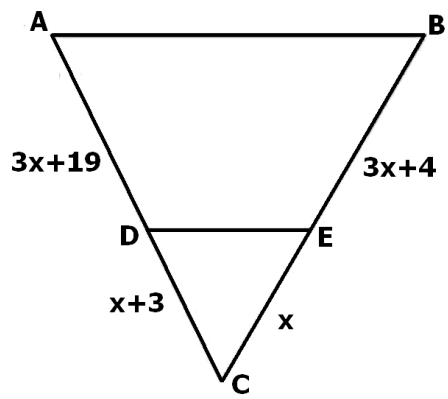
58. D is a point on side QR of ΔPQR such that PD ⊥ QR. Will it be correct to say that ΔPQD ~ Δ RPD? Why?

59. Is it true to say that if in two triangles, an angle of one triangle is equal to an angle of another triangle and two sides of one triangle are proportional to the two sides of the other triangle, then the triangles are similar? Give reasons for your answer

60. Legs (sides other than the hypotenuse) of a right triangle are of lengths 16cm and 8 cm. Find the length of the side of the largest square that can be inscribed in the triangle.

61. In the below Figure, ∠D = ∠E and $\frac{AD}{DB} = \frac{AE}{EC}$. Prove that BAC is an isosceles triangle.





62. Find the value of *x* for which DE || AB in the above right sided Figure.

63. In a Δ PQR, $PR^2 - PQ^2 = QR^2$ and M is a point on side PR such that QM ⊥ PR. Prove that QM^2 = PM × MR.

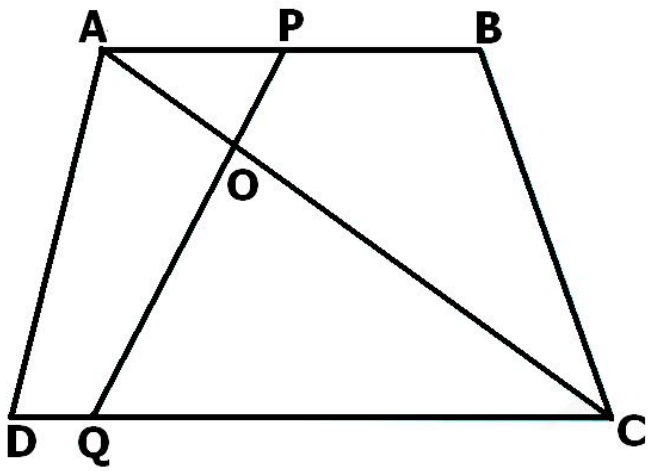
64. Hypotenuse of a right triangle is 25 cm and out of the remaining two sides, one is longer than the other by 5 cm. Find the lengths of the other two sides.

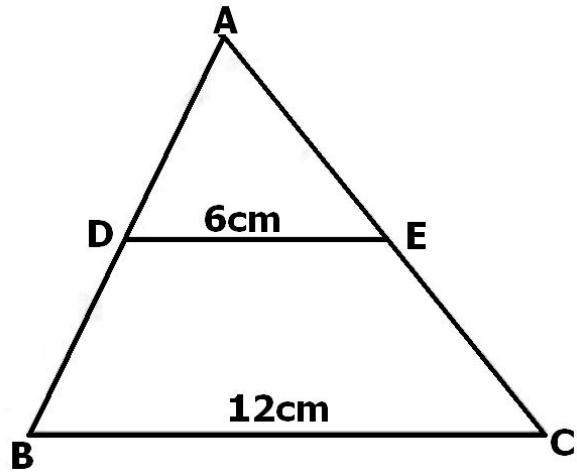
65. Diagonals of a trapezium PQRS intersect each other at the point O, PQ || RS and PQ = 3 RS. Find the ratio of the areas of triangles POQ and ROS.

66. Find the altitude of an equilateral triangle of side 8 cm.

67. If Δ ABC ~ Δ DEF, AB = 4 cm, DE = 6 cm, EF = 9 cm and FD = 12 cm, find the perimeter of Δ ABC.

68. In the below figure, if AB|| DC and AC and PQ intersect each other at the point O, prove that OA . CQ = OC . AP.





69. In the above right sided Figure, if DE || BC, find the ratio of ar (ADE) and ar (DECB).

70. ABCD is a trapezium in which AB || DC and P and Q are points on AD and BC, respectively such that PQ || DC. If PD = 18 cm, BQ = 35 cm and QC = 15 cm, find AD.

71. Corresponding sides of two similar triangles are in the ratio of 2 : 3. If the area of the smaller triangle is 48 cm^2, find the area of the larger triangle.

72. In a triangle PQR, N is a point on PR such that Q N $\perp$ PR . If PN. NR = QN^2, prove that $\angle PQR = 90°$.

73. A 15 metres high tower casts a shadow 24 metres long at a certain time and at the same time, a telephone pole casts a shadow 16 metres long. Find the height of the telephone pole.

74. Areas of two similar triangles are 36 cm^2 and 100 cm^2. If the length of a side of the larger triangle is 20 cm, find the length of the corresponding side of the smaller triangle.

75. Foot of a 10 m long ladder leaning against a vertical wall is 6 m away from the base of the wall. Find the height of the point on the wall where the top of the ladder reaches.

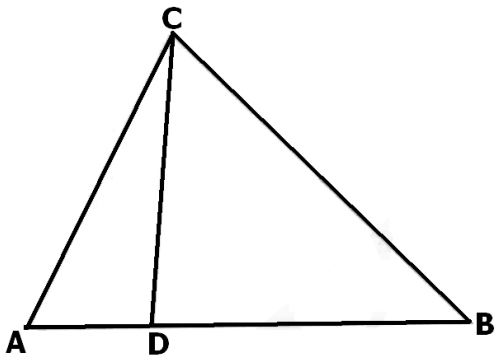
76. An aeroplane leaves an Airport and flies due North at 300 km/h. At the same time, another aeroplane leaves the same Airport and flies due West at 400 km/h. How far apart the two aeroplanes would be after $1\frac{1}{2}$ hours?

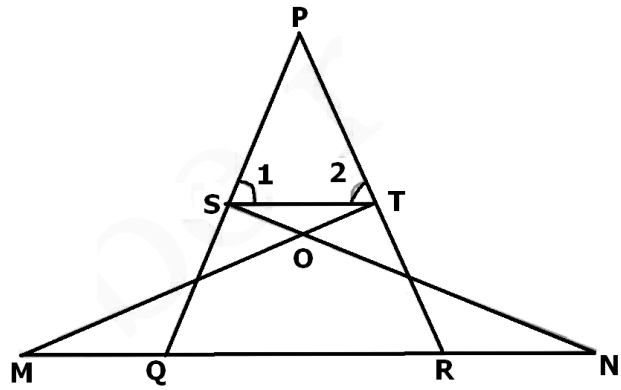
77. It is given that Δ ABC ~ Δ EDF such that AB = 5 cm, AC = 7 cm, DF= 15 cm and DE = 12 cm. Find the lengths of the remaining sides of the triangles.

78. A 5 m long ladder is placed leaning towards a vertical wall such that it reaches the wall at a point 4 m high. If the foot of the ladder is moved 1.6 m towards the wall, then find the distance by which the top of the ladder would slide upwards on the wall.

79. In a triangle PQR, PD $\perp$ QR such that D lies on QR . If PQ = a, PR = b, QD = c and DR = d, prove that $(a + b)(a - b) = (c + d)(c - d)$.

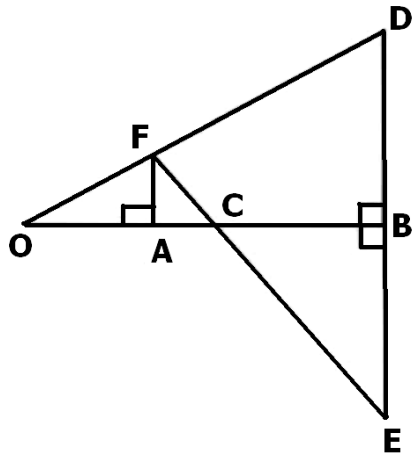
80. In the below Figure, if ∠ACB = ∠CDA, AC = 8 cm and AD = 3 cm, find BD.

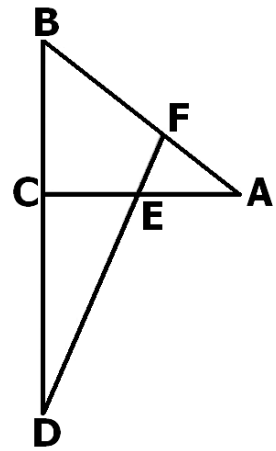




81. In the above right sided Figure, if ∠1 =∠2 and Δ NSQ ≅ Δ MTR, then prove that Δ PTS ~ Δ PRQ.

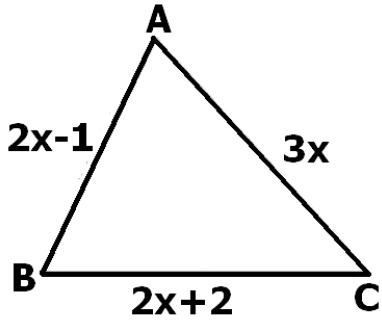
82. In the below Figure, OB is the perpendicular bisector of the line segment DE, FA ⊥ OB and F E intersects OB at the point C. Prove that $\frac{1}{OA}+\frac{1}{OB}=\frac{2}{OC}$

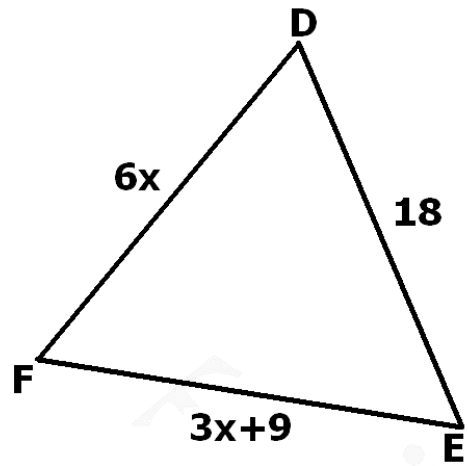




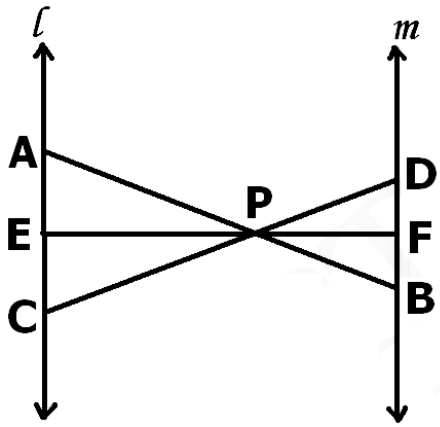
83. In the above right sided figure, line segment DF intersect the side AC of a triangle ABC at the point E such that E is the mid-point of CA and ∠AEF = ∠AFE. Prove that $\frac{BD}{CD}=\frac{BF}{CE}$.

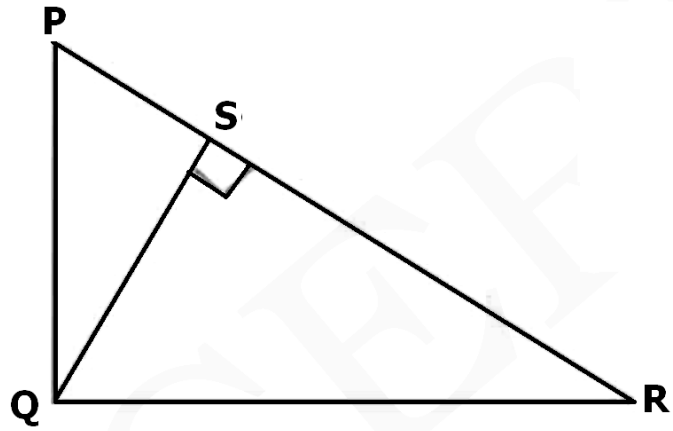
84. In the below figure, if Δ ABC ~ Δ DEF and their sides are of lengths (in cm) as marked along them, then find the lengths of the sides of each triangle.





85. In the below figure, l || m and line segments AB, CD and EF are concurrent at point P. Prove that $\frac{AE}{BF} = \frac{AC}{BD} = \frac{CE}{FD}$.

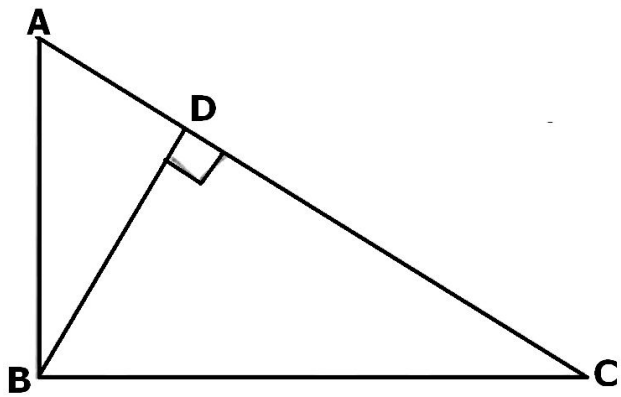


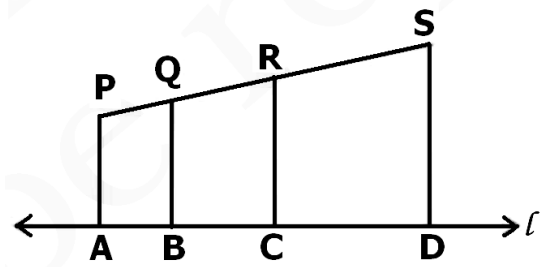


86. In the above right sided figure, PQR is a right triangle right angled at Q and QS ⊥ PR . If PQ = 6 cm and PS = 4 cm, find QS, RS and QR.

87. For going to a city B from city A, there is a route via city C such that AC⊥CB, AC = 2 x km and CB = 2 (x + 7) km. It is proposed to construct a 26 km highway which directly connects the two cities A and B. Find how much distance will be saved in reaching city B from city A after the construction of the highway.

88. In the below figure, ABC is a triangle right angled at B and BD ⊥ AC. If AD = 4 cm and CD = 5 cm, find BD and AB.





89. In the above right sided figure, PA, QB, RC and SD are all perpendiculars to a line l, AB = 6 cm, BC = 9 cm, CD = 12 cm and SP = 36 cm. Find PQ, QR and RS.

90. In a quadrilateral ABCD, ∠A = ∠D = 90°. Prove that $AC^2 + BD^2 = AD^2 + BC^2 + 2.CD.AB$

91. A flag pole 18 m high casts a shadow 9.6 m long. Find the distance of the top of the pole from the far end of the shadow.

92. A street light bulb is fixed on a pole 6 m above the level of the street. If a woman of height 1.5 m casts a shadow of 3m, find how far she is away from the base of the pole.

93. O is the point of intersection of the diagonals AC and BD of a trapezium ABCD with AB || DC. Through O, a line segment PQ is drawn parallel to AB meeting AD in P and BC in Q. Prove that PO = QO.

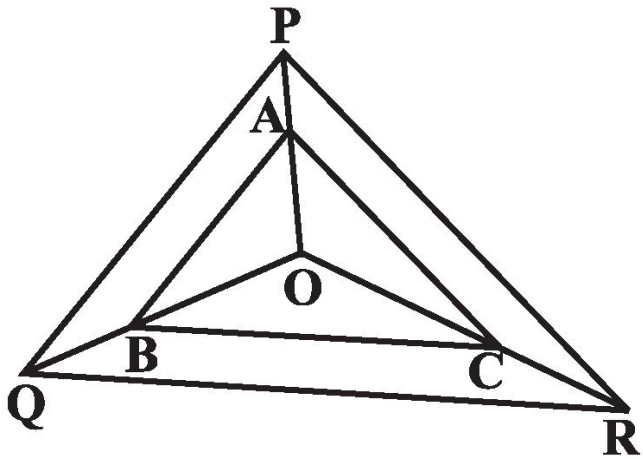
94. Prove that the internal bisector of an angle of a triangle divides the opposite side in the ratio of the sides containing the angle.

95. Prove that the area of the semicircle drawn on the hypotenuse of a right angled triangle is equal to the sum of the areas of the semicircles drawn on the other two sides of the triangle.

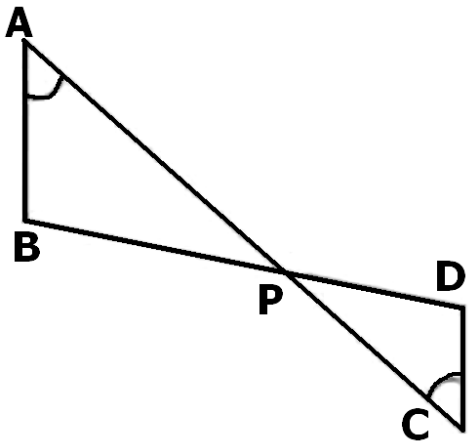
96. Using Thales theorem, prove that a line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side.

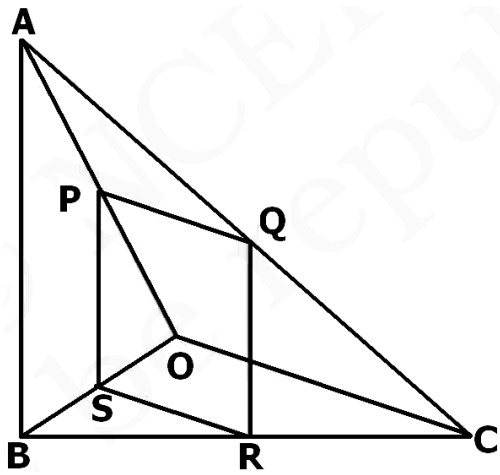
97. Using Converse of Thales theorem, prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side.

98. In the below figure A, B and C are points on OP, OQ and OR respectively such that AB || PQ and AC || PR. Show that BC || QR.



99. In the below figure, if ∠A = ∠C, AB = 6 cm, BP = 15 cm, AP = 12 cm and CP = 4 cm, then find the lengths of PD and CD.





100. In the above right sided figure, if PQRS is a parallelogram, AB || PS and PQ || OC, then prove that OC || SR.